Lagrangian Duality

We now use the Lagrangian to construct a new optimization problem in the multipliers. We will first obtain a lower bound on the original objective, then ask how large that bound can be. This construction will explain both what a dual formulation offers and what must still be checked before a dual solution can be used to solve the original problem.

Learning objectives

By the end of this chapter, you should be able to:

- derive the dual function and prove weak duality;
- use a primal–dual gap as a global optimality certificate;
- prove concavity of the dual function and identify its effective domain;
- distinguish zero duality gap, attainment, and primal recovery;
- classify the possible finite and infinite primal–dual values;
- distinguish nonuniqueness, nonattainment, and unboundedness at both levels;
- identify possible nonsmoothness of the dual function; and
- construct and solve a dual problem in a concrete example.

18.1 From primal iterations to a dual problem

18.1.1 The variables in the KKT conditions

To fix the notation, consider

$$\begin{aligned} p^* := \inf_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \\ & \boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}, \end{aligned} \tag{18.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$, $\boldsymbol{g} : \mathbb{R}^n \to \mathbb{R}^m$, and $\boldsymbol{h} : \mathbb{R}^n \to \mathbb{R}^p$. The feasible set is

$$\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0},\ \boldsymbol{h}(\boldsymbol{x}) \le \boldsymbol{0}\}.$$

The value p^* may be an infimum that is not attained. As usual, $p^* = +\infty$ if $\mathcal{S} = \varnothing$, and $p^* = -\infty$ if the objective is unbounded below on $\mathcal{S}$. When discussing optimal solutions and numerical gaps, we will explicitly require finite values.

Attach unrestricted multipliers $\boldsymbol{\lambda} \in \mathbb{R}^m$ to the equalities and multipliers $\boldsymbol{\nu} \in \mathbb{R}^p$ to the inequalities. The Lagrangian is

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) + \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x}). \tag{18.2}$$

When the data are differentiable, the KKT conditions are

$$\begin{aligned} \nabla_{\boldsymbol{x}} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}) &= \mathbf{0}, && \text{stationarity},\\ \boldsymbol{g}(\boldsymbol{x}) = \mathbf{0}, \qquad \boldsymbol{h}(\boldsymbol{x}) &\le \mathbf{0}, && \text{primal feasibility},\\ \boldsymbol{\nu} &\ge \mathbf{0}, && \text{dual feasibility},\\ \nu_j h_j(\boldsymbol{x}) &= 0 \quad (j=1,\dots,p), && \text{complementary slackness}. \end{aligned} \tag{18.3}$$

Their necessity and sufficiency were studied in Chapters 12 to 14. We will use the same multiplier signs here.

The algorithms in Chapters 15 to 17 were introduced by constructing subproblems for $\boldsymbol{x}$. This is a natural starting point: the coordinates of $\boldsymbol{x}$ represent the decisions in the original model. QP and ALM change the objective, LN and SQP use local KKT or quadratic models, and barrier methods put a wall near inequality boundaries. These methods also handle multipliers, but not all in the same way. QP and barrier methods provide implicit estimates; ALM updates its multipliers explicitly, and LN/SQP compute multiplier updates along with the primal step.

The Lagrangian contains both $\boldsymbol{x}$ and $(\boldsymbol{\lambda},\boldsymbol{\nu})$. We call these the *primal variables* and *dual variables*, respectively. To see whether the second group can be used as the main optimization variables, we need a scalar objective depending only on the multipliers. We will obtain it by eliminating $\boldsymbol{x}$ through a global infimum.

18.1.2 Weak duality

Fix a primal feasible $\boldsymbol{x}$ and multipliers with $\boldsymbol{\nu} \ge \mathbf{0}$. The equality terms vanish, and each inequality term is nonpositive. Therefore

$$L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}) = f(\boldsymbol{x}) + \sum_{j=1}^{p} \nu_j h_j(\boldsymbol{x}) \le f(\boldsymbol{x}).$$

This explains the sign restriction on $\boldsymbol{\nu}$: it makes every feasible primal objective an upper bound on the corresponding Lagrangian value. No sign restriction is needed for $\boldsymbol{\lambda}$.

The displayed lower bound still depends on $\boldsymbol{x}$. To remove that dependence, minimize the Lagrangian over all primal vectors, including those that violate the original constraints.

Definition 18.1 (Dual function and dual problem)**.** The *dual function* is

$$\theta(\boldsymbol{\lambda},\boldsymbol{\nu}) \coloneqq \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x},\boldsymbol{\lambda},\boldsymbol{\nu}). \tag{18.4}$$

The associated *Lagrangian dual problem* seeks the largest such bound:

$$d^* \coloneqq \sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\,\boldsymbol{\nu}\ge\mathbf{0}} \theta(\boldsymbol{\lambda},\boldsymbol{\nu}). \tag{18.5}$$

The function L, and hence θ, can be defined for every multiplier pair in $\mathbb{R}^m \times \mathbb{R}^p$. The condition $\boldsymbol{\nu} \ge \mathbf{0}$ is the sign restriction in the dual problem. We use "dual feasible" for multipliers satisfying this sign restriction; a finite dual value will be required whenever we use the point as a numerical certificate. The notation inf matters: the Lagrangian may have no minimizer, or its infimum may be $-\infty$. Because the data are real-valued on the nonempty set $\mathbb{R}^n$, the infimum cannot be $+\infty$.

Theorem 18.1 (Weak duality)**.** *For every primal feasible $\boldsymbol{x}$ and dual feasible $(\boldsymbol{\lambda}, \boldsymbol{\nu})$,*

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f(\boldsymbol{x}). \tag{18.6}$$

Consequently $d^* \leq p^*$.

Proof. The first inequality follows from the definition of an infimum; the second was obtained from feasibility and the multiplier signs. Keeping the multipliers fixed and taking the infimum over $\boldsymbol{x} \in \mathcal{S}$ gives

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \inf_{\boldsymbol{x} \in \mathcal{S}} f(\boldsymbol{x}) = p^*.$$

Now take the supremum over $\boldsymbol{\lambda} \in \mathbb{R}^m$ and $\boldsymbol{\nu} \geq \mathbf{0}$ to obtain $d^* \leq p^*$. If $\mathcal{S}$ is empty, the last inequality is automatic because $p^* = +\infty$. If $p^* = -\infty$, the pointwise bound forces every dual-feasible value to equal $-\infty$, and therefore $d^* = -\infty$. □

18.1.3 Duality gap

We will distinguish the difference between optimal values from a gap that can be evaluated at a candidate pair.

Definition 18.2 (Optimal duality gap)**.** When $p^*, d^* \in \mathbb{R}$, the *optimal duality gap* is $p^* - d^* \geq 0$. We say that *strong duality* holds when this gap is zero, that is, when $p^* = d^*$. This definition concerns values; it does not assert that either optimum is attained.

Let $\boldsymbol{x}$ be primal feasible and let $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ be dual feasible with $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$ finite. Combining the definitions of the two optimal values with weak duality gives

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq d^* \leq p^* \leq f(\boldsymbol{x}). \tag{18.7}$$

In particular, p^* and d^* are finite in this situation. Define the *primal–dual gap at the candidate pair* by

$$G(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Subtract the chain from $f(\boldsymbol{x})$ to obtain

$$0 \leq f(\boldsymbol{x}) - p^* \leq f(\boldsymbol{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}). \tag{18.8}$$

The quantity on the right is G, which can be computed without knowing p^*. To see the other errors that it bounds, insert the two optimal values between its endpoints:

$$G = \underbrace{f(\boldsymbol{x}) - p^*}_{\text{primal objective error}} + \underbrace{p^* - d^*}_{\text{optimal duality gap}} + \underbrace{d^* - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu})}_{\text{dual objective error}}. \tag{18.9}$$

Each term is nonnegative by Equation (18.7), and therefore

$$0 \leq \max\{f(\boldsymbol{x}) - p^*,\ p^* - d^*,\ d^* - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu})\} \leq G.$$

Thus a candidate gap of at most ε_{out} certifies that both objective errors, and also the optimal duality gap, are at most ε_{out}. The zero case is particularly useful.

Corollary 18.2 (Matching-value certificate)**.** *If $\boldsymbol{x}$ is primal feasible, $\boldsymbol{\nu} \geq \mathbf{0}$, and $f(\boldsymbol{x}) = \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathbb{R}$, then $\boldsymbol{x}$ and $(\boldsymbol{\lambda}, \boldsymbol{\nu})$ solve the primal and dual problems, respectively.*

Proof. The endpoints of Equation (18.7) agree, so every inequality in the chain is an equality. □

When strong duality holds and both optimal values are attained, any primal optimizer and dual optimizer supply such a certificate. Strong duality without attainment does not supply a candidate pair. Chapter 19 develops the connection with KKT conditions and global saddle points.

This certificate compares objective values with a bound valid at *every* feasible point. It does not require derivatives or a local optimality test. By comparison, KKT stationarity by itself need not distinguish a nonconvex minimum from a maximum or saddle point. The difficulty is that evaluating θ requires a global Lagrangian infimum, and a matching bound may not exist even when the original problem has an optimal solution.

To test whether a zero gap is automatic, consider the smooth scalar problem

$$\min_{x\in\mathbb{R}}(x^2-1)^2 \qquad \text{subject to} \qquad x=0. \tag{18.10}$$

The feasible set contains only 0, so the primal optimum is attained and $p^*=1$. Its Lagrangian is

$$L(x,\lambda)=(x^2-1)^2+\lambda x.$$

For every fixed λ, the quartic term dominates the linear term as $|x|\to\infty$, so the Lagrangian has a finite global minimum. We do not need its full formula to solve the dual. At $\lambda=0$,

$$\theta(0)=\min_x(x^2-1)^2=0.$$

For arbitrary λ, testing the two points $x=1$ and $x=-1$ gives

$$\theta(\lambda)\le\min\{L(1,\lambda),L(-1,\lambda)\}=\min\{\lambda,-\lambda\}=-|\lambda|\le 0.$$

Thus $\lambda^*=0$ maximizes the dual function, and

$$d^*=0<1=p^*.$$

Both problems attain their optimal values, yet the gap is positive. At the primal optimizer, $f'(0)=0$ and the equality gradient is 1, so $(x^*,\lambda^*)=(0,0)$ also satisfies KKT. Nevertheless $x^*=0$ does not minimize $L(\cdot,0)$ globally. This illustrates exactly why KKT and a matching-value certificate are different for nonconvex problems.

18.1.4 Dual form and concavity

The dual problem is

$$\sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\,\boldsymbol{\nu}\ge\mathbf{0}}\underbrace{\inf_{\boldsymbol{x}\in\mathbb{R}^n}\left\{f(\boldsymbol{x})+\boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{g}(\boldsymbol{x})+\boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{h}(\boldsymbol{x})\right\}}_{\theta(\boldsymbol{\lambda},\boldsymbol{\nu})}.$$

Its explicit constraints are only sign restrictions. There is one scalar dual variable for each scalar constraint, so the dual has $m+p$ variables, compared with n primal variables. When a large number of decisions are coupled by a few constraints, this can greatly reduce the outer dimension. There is no general inequality between n and $m+p$, however: a problem may have more constraints than variables, especially when some constraints are redundant.

We next examine the objective. Fixing $\boldsymbol{x}$ leaves an affine function of the multipliers. Taking its infimum over $\boldsymbol{x}$ preserves a useful one-sided curvature property.

Theorem 18.3 (Concavity of the dual function)**.** *The function θ in Equation (18.4) is concave, in the extended-real sense, regardless of convexity of the primal problem.*

Proof. Write $\boldsymbol{w} = (\boldsymbol{\lambda}, \boldsymbol{\nu})$, take $0 < t < 1$, and first suppose $\theta(\boldsymbol{w}_1)$ and $\theta(\boldsymbol{w}_2)$ are finite. For every $\boldsymbol{x}$,

$$\begin{aligned} L(\boldsymbol{x}, t\boldsymbol{w}_1 + (1-t)\boldsymbol{w}_2) &= tL(\boldsymbol{x}, \boldsymbol{w}_1) + (1-t)L(\boldsymbol{x}, \boldsymbol{w}_2) \\ &\geq t\theta(\boldsymbol{w}_1) + (1-t)\theta(\boldsymbol{w}_2). \end{aligned}$$

Taking the infimum over $\boldsymbol{x}$ on the left gives

$$\theta(t\boldsymbol{w}_1 + (1-t)\boldsymbol{w}_2) \geq t\theta(\boldsymbol{w}_1) + (1-t)\theta(\boldsymbol{w}_2).$$

If either endpoint value equals $-\infty$, the right-hand side is $-\infty$, so the inequality remains valid. The endpoint cases $t = 0, 1$ are identities. □

Maximizing this concave function over the convex sign-restricted multiplier set is a convex optimization problem, equivalently a minimization of the convex function $-\theta$. This statement describes the outer problem's geometry. It does not assert that θ is finite, differentiable, or inexpensive to evaluate.

18.1.5 Example: a complete dual construction and solution

We now apply the preceding steps to a problem for which the inner infimum, outer supremum, and primal recovery can all be computed explicitly:

$$\min_{(x_1,x_2)\in\mathbb{R}^2} \frac{1}{2}(x_1^2 + x_2^2) \qquad \text{subject to} \qquad x_1 + x_2 = 4. \tag{18.11}$$

The equality couples the two coordinates. Introduce an unrestricted scalar multiplier λ and group terms by coordinate:

$$\begin{aligned} L(\boldsymbol{x}, \lambda) &= \frac{1}{2}(x_1^2 + x_2^2) + \lambda(x_1 + x_2 - 4) \\ &= \left(\frac{1}{2}x_1^2 + \lambda x_1\right) + \left(\frac{1}{2}x_2^2 + \lambda x_2\right) - 4\lambda. \end{aligned}$$

For fixed λ, each bracket is a strictly convex scalar quadratic. Completing the square gives

$$\frac{1}{2}x_i^2 + \lambda x_i = \frac{1}{2}(x_i + \lambda)^2 - \frac{1}{2}\lambda^2.$$

Each minimum is attained uniquely at $x_i(\lambda) = -\lambda$, so

$$\theta(\lambda) = -\lambda^2 - 4\lambda = 4 - (\lambda + 2)^2.$$

The inner infimum is finite for every $\lambda \in \mathbb{R}$. The dual problem is the unconstrained scalar maximization

$$\max_{\lambda\in\mathbb{R}} \{4 - (\lambda + 2)^2\},$$

whose unique solution is $\lambda^* = -2$ and value is $d^* = 4$.

To recover the primal point, substitute this multiplier into the minimizing response: $\boldsymbol{x}(\lambda^*) = (2, 2)^\mathsf{T}$. Check the equality rather than assume it: $2 + 2 = 4$. Its objective value is 4, matching the dual value, so Equation (18.8) certifies global optimality. Uniqueness follows from

strict convexity of the objective on the feasible line. As an independent direct calculation, substituting $x_2 = 4 - x_1$ gives

$$\frac{1}{2}(x_1^2 + (4 - x_1)^2) = (x_1 - 2)^2 + 4,$$

which confirms the same minimizer and value.

The construction also yields a certificate before either candidate is optimal. Choose the feasible point $\widehat{\boldsymbol{x}} = (3, 1)^\mathsf{T}$ and multiplier $\widehat{\lambda} = -1$. Then

$$f(\widehat{\boldsymbol{x}}) = 5, \qquad \theta(\widehat{\lambda}) = 3, \qquad 3 \le p^* \le 5,$$

so the objective error of $\widehat{\boldsymbol{x}}$ is at most 2. The inner response at $\widehat{\lambda} = -1$ is instead $(1, 1)^\mathsf{T}$. Its original objective value is 1, but this is not a primal upper bound: the point violates the resource equality. The lower bound 3 comes from its *Lagrangian* value.

18.1.6 Example: more variables, the same outer dimension

To see the dimensional benefit more clearly, replace the two costs by

$$\min_{\boldsymbol{x} \in \mathbb{R}^N} \frac{1}{2} \sum_{i=1}^{N} a_i x_i^2 \qquad \text{subject to} \qquad \sum_{i=1}^{N} x_i = b, \qquad a_i > 0.$$

For fixed λ, the i-th minimizer satisfies $a_i x_i + \lambda = 0$, so $x_i(\lambda) = -\lambda / a_i$. Summing the minimum values gives

$$\theta(\lambda) = -\frac{\lambda^2}{2} \sum_{i=1}^{N} \frac{1}{a_i} - b\lambda.$$

The positive number $s = \sum_i 1/a_i$ allows us to write this as $b^2/(2s) - \frac{s}{2}(\lambda + b/s)^2$. Hence

$$\lambda^* = -\frac{b}{\sum_i 1/a_i}, \qquad x_i^* = \frac{b/a_i}{\sum_j 1/a_j}, \qquad p^* = d^* = \frac{b^2}{2\sum_i 1/a_i}.$$

Again, $\sum_i x_i^* = b$ verifies recovery. One multiplier coordinates N independent inner calculations; it does not eliminate the work in those calculations.

We can also interpret the multiplier in this example. Regard the last value as a function $p(b) = b^2/(2s)$. Differentiation gives

$$p'(b) = \frac{b}{s} = -\lambda^*(b).$$

The minus sign comes from writing the equality as $\sum_i x_i - b = 0$. This is an example of a sensitivity interpretation: the multiplier describes the marginal change in optimal value as the required resource changes.

18.2 Difficulties in dual optimization

The simple multiplier constraints, possible reduction in dimension, and concavity can make the dual look easier than the primal. To assess this impression, we must examine the inner infimum and the passage back to a primal answer. The following examples isolate the relevant issues.

18.2.1 Infima, minimizers, and the effective domain

We can exhibit three different inner behaviors without changing the primal problem. Consider

$$\min_{x\in\mathbb{R}} e^x \qquad \text{subject to} \qquad 1 - e^x \le 0. \tag{18.12}$$

The constraint is equivalent to $x \ge 0$, so the unique primal optimizer is $x^* = 0$, with $p^* = 1$. For a multiplier ν, form

$$L(x, \nu) = e^x + \nu(1 - e^x) = (1 - \nu)e^x + \nu.$$

We obtain the infimum by inspecting the coefficient of e^x:

- If $\nu < 1$, the coefficient is positive. The Lagrangian is strictly larger than ν at every finite x, and approaches ν as $x \to -\infty$. Its infimum is finite but not attained.
- If $\nu = 1$, the Lagrangian is the constant 1. Every $x \in \mathbb{R}$ is an inner minimizer.
- If $\nu > 1$, the coefficient is negative. Sending $x \to +\infty$ makes the Lagrangian tend to $-\infty$.

Consequently, on the entire multiplier line,

$$\theta(\nu) = \begin{cases} \nu, & \nu \le 1, \\ -\infty, & \nu > 1. \end{cases} \tag{18.13}$$

The dual problem additionally requires $\nu \ge 0$. Its maximum is 1, attained uniquely at $\nu^* = 1$, and matches the primal value. Thus inner nonattainment at some multipliers can coexist with an attained dual optimum. No convexity of the primal formulation was needed for this calculation; in particular, the displayed inequality function $1 - e^x$ is not convex.

The finite and infinite cases motivate the *effective domain*

$$\operatorname{dom}\theta = \{(\boldsymbol{\lambda}, \boldsymbol{\nu}) : \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) > -\infty\}. \tag{18.14}$$

To obtain a finite lower bound, a dual point must belong to this domain as well as satisfy $\boldsymbol{\nu} \ge \mathbf{0}$. Write $\mathcal{F}_D = \operatorname{dom}\theta \cap (\mathbb{R}^m \times \mathbb{R}^p_+)$ for the set of such finite-value multiplier pairs. Removing the $-\infty$-valued points does not change the dual supremum, so

$$d^* = \sup_{(\boldsymbol{\lambda},\boldsymbol{\nu})\in\mathcal{F}_D} \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}), \qquad \sup\varnothing = -\infty.$$

This remains a supremum; attainment must be checked separately. The explicit constraints of the dual are simple, but the inner boundedness requirement can hide a less transparent restriction. In Equation (18.13),

$$\operatorname{dom}\theta = (-\infty, 1], \qquad \mathcal{F}_D = [0, 1].$$

The distinction keeps track of the required signs. It is not enough to solve $\nabla_{\boldsymbol{x}} L = \mathbf{0}$: any stationary candidate must still be checked against the global infimum.

With this finite-value formulation, we can classify the possible primal and dual optimal values before examining attainment. If a primal feasible point $\bar{\boldsymbol{x}}$ exists, then $p^* \le f(\bar{\boldsymbol{x}}) < +\infty$, because f is finite everywhere. Together with the convention for an empty feasible set, this gives

$$p^* = +\infty \quad \Longleftrightarrow \quad \mathcal{S} = \varnothing.$$

On the dual side, a point in $\mathcal{F}_D$ supplies a finite lower bound on d^*. If $\mathcal{F}_D$ is empty, every sign-admissible multiplier instead gives $-\infty$, and so does their supremum. Hence

$$d^* = -\infty \quad \Longleftrightarrow \quad \mathcal{F}_D = \varnothing. \tag{18.15}$$

The sign-admissible set itself always contains the zero multipliers. An empty $\mathcal{F}_D$ means that none of them gives a finite bound; it is the *finite-value* dual formulation that has no feasible point.

Each optimal value can be $-\infty$, finite, or $+\infty$. Weak duality rules out a finite d^* when $p^* = -\infty$, and rules out $d^* = +\infty$ when p^* is finite or $-\infty$. These eliminate three of the nine combinations, leaving the following table. The middle cell still requires $d^* \le p^*$.

	$p^* = +\infty$	$p^* \in \mathbb{R}$	$p^* = -\infty$
$d^* = +\infty$	$\mathcal{S} = \varnothing$	Impossible	Impossible
$d^* \in \mathbb{R}$	$\mathcal{S} = \varnothing$	$d^* \le p^*$	Impossible
$d^* = -\infty$	$\mathcal{S} = \varnothing,\ \mathcal{F}_D = \varnothing$	$\mathcal{F}_D = \varnothing$	$\mathcal{F}_D = \varnothing$

The finite–finite entry has already appeared: the resource and exponential problems have equal optimal values, whereas Equation (18.10) has a positive gap. We now use examples to fill the other five entries, starting with primal unboundedness.

Suppose $p^* = -\infty$, and choose primal feasible points $\boldsymbol{x}^{(k)}$ such that $f(\boldsymbol{x}^{(k)}) \to -\infty$. For every fixed sign-admissible multiplier pair, weak duality gives

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \le f(\boldsymbol{x}^{(k)}) \quad \text{for every } k.$$

The only possible dual value is therefore $-\infty$. Since the multiplier pair was arbitrary,

$$p^* = -\infty \quad \Longrightarrow \quad \mathcal{F}_D = \varnothing, \qquad d^* = -\infty. \tag{18.16}$$

The conclusion concerns $\mathcal{F}_D$, rather than the domain before the sign restrictions are imposed. For example, $\min -x$ subject to $-x \le 0$ is unbounded below, and $L(x, \nu) = -(1 + \nu)x$ has infimum $-\infty$ for every $\nu \ge 0$, although $\theta(-1) = 0$. Thus $\operatorname{dom}\theta$ itself need not be empty.

The converse of Equation (18.16) need not hold. To see this using the same constraint as Equation (18.10), write $q(x) = (x^2 - 1)^2$ and reverse the sign of its objective:

$$\min_{x \in \mathbb{R}} -q(x) \qquad \text{subject to} \qquad x = 0. \tag{18.17}$$

The feasible set is $\{0\}$, so the primal problem is bounded below, attains its minimum, and has $p^* = -1$. For any fixed multiplier,

$$L(x, \lambda) = -(x^2 - 1)^2 + \lambda x \longrightarrow -\infty \qquad \text{as } x \to +\infty.$$

The quartic dominates every fixed linear correction. Consequently

$$\theta(\lambda) = -\infty \quad \text{for every } \lambda \in \mathbb{R}, \qquad \operatorname{dom}\theta = \varnothing, \qquad d^* = -\infty.$$

Weak duality remains valid, but it supplies only the useless bound $-\infty \le -1$. The dual is a maximization problem whose objective is identically $-\infty$, rather than an ordinary finite-valued objective that becomes arbitrarily negative along only some multipliers.

There is no contradiction with boundedness of the constrained primal problem. The inner infimum removes the equality and explores points at which $-q(x)$ is arbitrarily negative. If instead f were bounded below on *all* of $\mathbb{R}^n$, then $\theta(\mathbf{0}, \mathbf{0}) = \inf_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x}) > -\infty$. The zero multipliers are dual feasible, so an empty effective domain would then be impossible.

The remaining three entries all have $p^* = +\infty$. We will realize each of them, beginning with a finite dual value. Take

$$\min_{x\in\mathbb{R}} 0 \qquad \text{subject to} \qquad e^x \le 0.$$

No finite x is feasible, so $p^* = +\infty$. For every $\nu \ge 0$, however,

$$\theta(\nu) = \inf_x \nu e^x = 0.$$

At $\nu = 0$ the inner objective is constant; at $\nu > 0$ the infimum is approached as $x \to -\infty$. Thus $d^* = 0$, and every nonnegative multiplier is optimal. Primal infeasibility alone does not force the dual objective to be unbounded above.

Next take

$$\min_{x\in\mathbb{R}} -x^4 \qquad \text{subject to} \qquad x^2 + 1 \le 0.$$

The primal is again infeasible. For any fixed $\nu \ge 0$, the negative quartic dominates $\nu(x^2+1)$ as $|x|$ grows:

$$\theta(\nu) = \inf_x \{-x^4 + \nu(x^2+1)\} = -\infty.$$

Here $\mathcal{F}_D = \varnothing$, giving $(p^*, d^*) = (+\infty, -\infty)$.

For the last entry, keep this infeasible constraint but replace the objective by x^2:

$$\min_{x\in\mathbb{R}} x^2 \qquad \text{subject to} \qquad x^2 + 1 \le 0.$$

The primal problem has no feasible point. For every $\nu \ge 0$,

$$L(x, \nu) = (1+\nu)x^2 + \nu, \qquad \theta(\nu) = \nu.$$

The inner minimum is attained uniquely at $x = 0$, yet the resulting dual values grow without bound as $\nu \to +\infty$. Thus $d^* = +\infty$, consistent with $p^* = +\infty$.

More generally, if $d^* = +\infty$, a primal feasible point $\boldsymbol{x}$ would give the contradiction $+\infty = d^* \le f(\boldsymbol{x}) < +\infty$. Thus a dual unbounded above certifies *primal infeasibility*, not merely failure to attain a primal minimum.

We have now realized all six entries allowed by weak duality. The table does not determine uniqueness or attainment. For example, $\inf_{x\in\mathbb{R}} e^x$ has a nonempty feasible set and finite value 0, but no optimizer. Such nonattainment is different from an empty feasible set. We next examine the outer supremum before considering primal recovery and smoothness.

18.2.2 The outer supremum and its optimizers

The preceding examples distinguish the possible values of the outer supremum. We now ask whether a finite supremum is attained and whether its maximizer is unique. First record the regularity supplied by the infimum construction.

Proposition 18.4 (Effective domain and upper semicontinuity)**.** *The effective domain*

of θ is convex. Moreover, the hypograph

$$\operatorname{hypo}\theta = \{(\boldsymbol{w}, t) : t \le \theta(\boldsymbol{w})\}$$

is closed, so θ is upper semicontinuous as an extended-real concave function.

Proof. If $\theta(\boldsymbol{w}_1)$ and $\theta(\boldsymbol{w}_2)$ are finite, concavity bounds the value at any convex combination below by their finite weighted average. Thus the combination is in $\operatorname{dom}\theta$. For the second claim, observe that

$$\operatorname{hypo}\theta = \bigcap_{\boldsymbol{x}\in\mathbb{R}^n} \{(\boldsymbol{w}, t) : t \le L(\boldsymbol{x}, \boldsymbol{w})\}.$$

For each fixed $\boldsymbol{x}$, $L(\boldsymbol{x}, \boldsymbol{w})$ is affine and continuous in $\boldsymbol{w}$, so the set in braces is a closed halfspace. Their intersection is closed. □

Closedness of the hypograph does not guarantee a dual maximizer: a maximizing sequence may escape to infinity.

For a finite maximum with infinitely many maximizers, consider

$$\min_{x\in\mathbb{R}} x^2 \qquad \text{subject to} \qquad x^2 \le 0.$$

The primal point is forced to be 0, so $p^* = 0$. For every $\nu \ge 0$,

$$L(x, \nu) = (1 + \nu)x^2, \qquad \theta(\nu) = 0.$$

Thus $d^* = 0$, and every nonnegative multiplier is a dual optimizer. Here the inner minimizer is unique for each multiplier, even though the outer maximizer is not unique. Inner and outer nonuniqueness are distinct questions.

For a finite supremum that is not attained, change the objective to x:

$$\min_{x\in\mathbb{R}} x \qquad \text{subject to} \qquad x^2 \le 0. \tag{18.18}$$

Again $x^* = 0$ and $p^* = 0$. At $\nu = 0$, the Lagrangian $L(x, 0) = x$ is unbounded below. For $\nu > 0$, completing the square gives

$$L(x, \nu) = x + \nu x^2 = \nu\left(x + \frac{1}{2\nu}\right)^2 - \frac{1}{4\nu},$$

so the unique inner minimizer is $x(\nu) = -1/(2\nu)$, and

$$\theta(\nu) = \begin{cases} -1/(4\nu), & \nu > 0, \\ -\infty, & \nu = 0. \end{cases} \tag{18.19}$$

Every finite feasible dual value is strictly negative, but $\theta(\nu) \uparrow 0$ as $\nu \to +\infty$. Thus $d^* = p^* = 0$, while no dual optimizer exists. In fact, no multiplier can satisfy KKT at $x^* = 0$, because stationarity would require $1 + 2\nu x^* = 1 = 0$. This supplies a concrete reason for the attainment qualification in Proposition 19.3.

The earlier example with objective x^2 and constraint $x^2 + 1 \le 0$ already shows an unbounded outer supremum, even though every inner minimizer is unique. A missing dual maximizer is a much weaker statement: Equation (18.18) has many finite dual values and a finite primal optimum. Likewise, a finite primal optimum need not supply any finite dual value, as Equation (18.17) already showed.

18.2.3 Recovering a primal optimizer

Even when the dual optimum is finite and attained, an inner minimizer need not be the desired primal answer. Return to Equation (18.12). The dual solution is $\nu^* = 1$, and every real x minimizes $L(\cdot, 1) = 1$. The point $x = -1$ is not primal feasible. The point $x = 1$ is feasible but still fails to solve the primal problem, whose minimum is 1 at $x = 0$.

To see what is missing, take sign-admissible multipliers $(\boldsymbol{\lambda}, \boldsymbol{\nu})$, with $\boldsymbol{\nu} \geq \mathbf{0}$, and let $\bar{\boldsymbol{x}}$ minimize $L(\cdot, \boldsymbol{\lambda}, \boldsymbol{\nu})$ globally. In order to turn its Lagrangian value into the original objective without increasing it, we need

$$\boldsymbol{g}(\bar{\boldsymbol{x}}) = \mathbf{0}, \quad \boldsymbol{h}(\bar{\boldsymbol{x}}) \leq \mathbf{0}, \qquad \nu_j h_j(\bar{\boldsymbol{x}}) = 0 \quad (j = 1, \ldots, p).$$

Then

$$f(\bar{\boldsymbol{x}}) = L(\bar{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}),$$

and Corollary 18.2 applies. In the exponential example, complementary slackness at $\nu = 1$ gives $1 - e^x = 0$, selecting $x = 0$. For equality-only problems, primal feasibility is the additional requirement; with inequalities, complementary slackness must also be checked. Inner nonuniqueness can therefore turn primal recovery into a separate selection problem.

18.2.4 A smooth Lagrangian can produce a nonsmooth dual

We now return to the quartic problem Equation (18.10). Its Lagrangian is smooth in (x, λ), and its dual function is finite for every λ. Nevertheless the two inner minimizers $x = 1$ and $x = -1$ at $\lambda = 0$ produce a corner in the dual. Indeed, the bounds already derived give

$$\theta(0) = 0, \qquad \theta(\lambda) \leq -|\lambda|.$$

If a derivative at 0 existed, the inequality for $\lambda > 0$ would force it to be at most -1, whereas the inequality for $\lambda < 0$, after division by a negative number, would force it to be at least 1. Hence θ is not differentiable at 0. See Figure 18.1 for an illustration.

For a more precise description, let x_λ be any global minimizer. Comparison with the test point having sign opposite to λ gives $\theta(\lambda) \leq -|\lambda|$. In particular, for $\lambda > 0$, x_λ cannot be nonnegative: every nonnegative x gives $(x^2 - 1)^2 + \lambda x \geq 0$, whereas the test point -1 gives $-\lambda < 0$. To justify the limit as $\lambda \downarrow 0$, restrict to $0 < \lambda \leq 1$ and write $q(x) = (x^2 - 1)^2$. The same comparison gives

$$q(x_\lambda) + \lambda x_\lambda \leq -\lambda \leq 0, \qquad 0 \leq q(x_\lambda) \leq \lambda |x_\lambda| \leq |x_\lambda|.$$

The quartic $q(x)$ eventually exceeds $|x|$, so the minimizers lie in one bounded interval independent of λ. The sharper bound $q(x_\lambda) \leq \lambda |x_\lambda|$ then gives $q(x_\lambda) \to 0$. Every limit point lies in $\{-1, 1\}$, and the negative sign forces $x_\lambda \to -1$. Since

$$\lambda x_\lambda \leq \theta(\lambda) \leq -\lambda \qquad (\lambda > 0),$$

division by λ gives the right derivative -1. Symmetry gives the left derivative 1. To evaluate the function away from zero, solve $4x(x^2 - 1) + \lambda = 0$ and compare the Lagrangian values at all real roots; coercivity ensures that a global minimizer is among them.

The mechanism is visible without solving the quartic explicitly. If $\bar{\boldsymbol{x}}$ minimizes $L(\cdot, \boldsymbol{w})$, then for any other multiplier pair $\widehat{\boldsymbol{w}} = (\widehat{\boldsymbol{\lambda}}, \widehat{\boldsymbol{\nu}})$,

$$\begin{aligned} \theta(\widehat{\boldsymbol{w}}) &\leq L(\bar{\boldsymbol{x}}, \widehat{\boldsymbol{w}}) \\ &= \theta(\boldsymbol{w}) + \boldsymbol{g}(\bar{\boldsymbol{x}})^{\mathsf{T}}(\widehat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}) + \boldsymbol{h}(\bar{\boldsymbol{x}})^{\mathsf{T}}(\widehat{\boldsymbol{\nu}} - \boldsymbol{\nu}). \end{aligned}$$

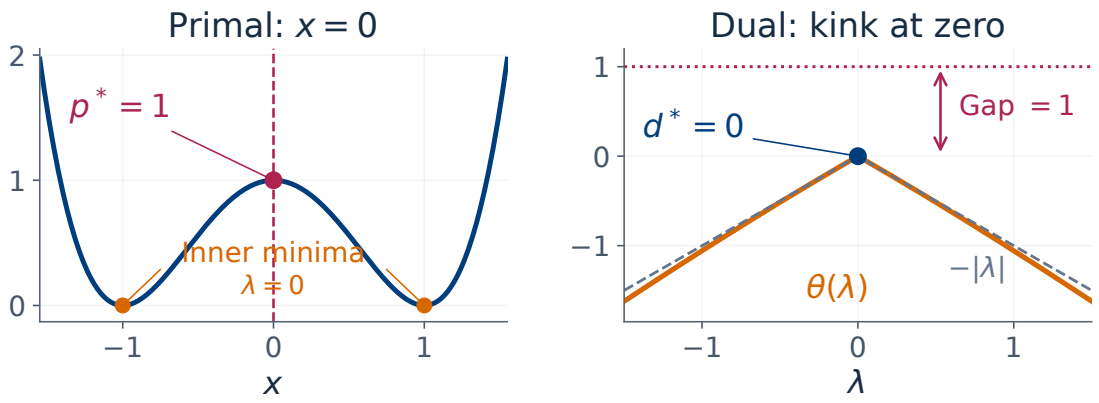


Figure 18.1: The quartic example has a finite concave dual with a corner at $\lambda = 0$. At that multiplier, the two inner minimizers have equality residuals 1 and -1, giving different supporting slopes. The primal value is 1, whereas the best dual value is 0.

Thus different minimizers can supply different affine upper supports to the same concave function. These residual vectors are called supergradients; Chapter 20 will use them to update the dual variables without assuming ordinary differentiability.

18.2.5 Approximate inner solutions and certified bounds

Fix dual-feasible multipliers $(\boldsymbol{\lambda}, \boldsymbol{\nu})$, with $\boldsymbol{\nu} \geq \mathbf{0}$. In a numerical method, an inner solver usually returns an approximation $\widetilde{\boldsymbol{x}}$. It is essential to keep the direction of the infimum inequality:

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}). \tag{18.20}$$

The computed value is an *upper* bound on the inner infimum. It need not be a lower bound on p^*. To check this in the resource example Equation (18.11), fix the optimal multiplier $\lambda = -2$. Completing the squares gives

$$L(\boldsymbol{x}, -2) = 4 + \frac{1}{2}(x_1 - 2)^2 + \frac{1}{2}(x_2 - 2)^2.$$

For $\widetilde{\boldsymbol{x}} = (2 + e, 2)^\mathsf{T}$ with any $e \neq 0$, the computed value is $4 + e^2/2 > 4 = p^*$. Even an arbitrarily small inner error can therefore invalidate the claimed exact lower bound.

Suppose an additional argument certifies an objective error bound

$$L(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq \delta.$$

Rearranging this inequality and using weak duality gives

$$L(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \delta \leq \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*. \tag{18.21}$$

The corrected value is therefore a valid lower bound. Without a certified δ, a small step or small gradient alone does not justify replacing θ by the computed inner objective in an optimality certificate.

For one useful way to obtain δ, suppose that $F(\boldsymbol{x}) = L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ is differentiable and globally strongly convex with known modulus $a > 0$. Its quadratic lower bound gives, for every $\boldsymbol{y}$,

$$F(\boldsymbol{y}) \geq F(\widetilde{\boldsymbol{x}}) + \nabla F(\widetilde{\boldsymbol{x}})^\mathsf{T}(\boldsymbol{y} - \widetilde{\boldsymbol{x}}) + \frac{a}{2}\left\|\boldsymbol{y} - \widetilde{\boldsymbol{x}}\right\|^2.$$

The right side is minimized at $\boldsymbol{y} - \widetilde{\boldsymbol{x}} = -\nabla F(\widetilde{\boldsymbol{x}})/a$. Taking the infimum on the left therefore yields

$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \geq F(\widetilde{\boldsymbol{x}}) - \frac{\|\nabla F(\widetilde{\boldsymbol{x}})\|^2}{2a}.$$

Thus $\delta = \|\nabla F(\widetilde{\boldsymbol{x}})\|^2/(2a)$ is valid in this setting. If an inner iteration uses $\|\nabla F(\widetilde{\boldsymbol{x}})\|_\infty \leq \varepsilon_{\text{in}}^{(k)}$, then $\delta \leq n(\varepsilon_{\text{in}}^{(k)})^2/(2a)$. The global curvature hypothesis is doing the work: this conclusion does not follow from an arbitrary local stationary solve of a nonconvex Lagrangian. For a primal feasible candidate $\bar{\boldsymbol{x}}$, the corrected value supplies the computable certificate

$$0 \leq f(\bar{\boldsymbol{x}}) - p^* \leq f(\bar{\boldsymbol{x}}) - (L(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu}) - \delta).$$

Thus the outer accuracy must account for a certified global inner error; a small gradient without the required curvature information does not suffice.

18.3 Supplement

18.3.1 Why a separable infimum splits

The calculation in the resource example extends to sums of independent blocks. We record the infimum argument because a block minimum need not be attained.

Proposition 18.5 (Infimum of a finite separable sum)**.** *Suppose $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N)$, with $\boldsymbol{x}_i \in \mathbb{R}^{n_i}$, and for fixed multipliers $\boldsymbol{w}$,*

$$L(\boldsymbol{x}, \boldsymbol{w}) = \sum_{i=1}^{N} L_i(\boldsymbol{x}_i, \boldsymbol{w}),$$

where each block function is real-valued. If $\theta_i(\boldsymbol{w}) = \inf_{\boldsymbol{x}_i \in \mathbb{R}^{n_i}} L_i(\boldsymbol{x}_i, \boldsymbol{w})$, then

$$\theta(\boldsymbol{w}) = \sum_{i=1}^{N} \theta_i(\boldsymbol{w}), \tag{18.22}$$

with a sum containing $-\infty$ interpreted as $-\infty$.

Proof. First suppose every block infimum is finite. For any $\boldsymbol{x}$, summing $L_i(\boldsymbol{x}_i, \boldsymbol{w}) \geq \theta_i(\boldsymbol{w})$ gives the lower bound $L(\boldsymbol{x}, \boldsymbol{w}) \geq \sum_i \theta_i(\boldsymbol{w})$. Conversely, for any $\varepsilon > 0$, choose each block so that

$$L_i(\boldsymbol{x}_i^\varepsilon, \boldsymbol{w}) \leq \theta_i(\boldsymbol{w}) + \frac{\varepsilon}{N}.$$

Their concatenation is a point of $\mathbb{R}^n$ whose total value is at most $\sum_i \theta_i(\boldsymbol{w}) + \varepsilon$. Letting $\varepsilon \downarrow 0$ proves equality. If one block infimum is $-\infty$, fix arbitrary values for all other blocks and make the chosen block arbitrarily negative. The total infimum is then $-\infty$. □

For a coupling equality $\sum_i \boldsymbol{r}_i(\boldsymbol{x}_i) = \boldsymbol{b}$ and objective $\sum_i f_i(\boldsymbol{x}_i)$, this gives

$$\theta(\boldsymbol{\lambda}) = -\boldsymbol{\lambda}^\mathsf{T}\boldsymbol{b} + \sum_i \inf_{\boldsymbol{x}_i \in \mathbb{R}^{n_i}} \{f_i(\boldsymbol{x}_i) + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{r}_i(\boldsymbol{x}_i)\}.$$

The blocks share the same multiplier but can be minimized independently. Their constraint residuals are then combined to inform a dual update. If the Lagrangian contains terms coupling different blocks, this decomposition is not available.

18.3.2 A quadratic calculation including singular Hessians

For an equality-constrained quadratic, the effective domain can be found explicitly from a range condition.

Proposition 18.6 (Dual function of an equality-constrained quadratic)**.** *Let $\boldsymbol{Q} \in \mathbb{R}^{n\times n}$ be symmetric positive semidefinite and consider*

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \qquad \textit{subject to} \qquad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}.$$

Put $\boldsymbol{a}(\boldsymbol{\lambda}) = \boldsymbol{c} + \boldsymbol{A}^{\mathsf{T}}\boldsymbol{\lambda}$. Then

$$\theta(\boldsymbol{\lambda}) = \begin{cases} -\frac{1}{2}\boldsymbol{a}(\boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{Q}^{\dagger}\boldsymbol{a}(\boldsymbol{\lambda}) - \boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}, & \boldsymbol{a}(\boldsymbol{\lambda}) \in \operatorname{Range}(\boldsymbol{Q}), \\ -\infty, & \boldsymbol{a}(\boldsymbol{\lambda}) \notin \operatorname{Range}(\boldsymbol{Q}), \end{cases} \tag{18.23}$$

where $\boldsymbol{Q}^{\dagger}$ is the Moore–Penrose pseudoinverse. If $\boldsymbol{Q}$ is positive definite, the range condition is automatic and $\boldsymbol{Q}^{\dagger} = \boldsymbol{Q}^{-1}$.

Proof. For fixed $\boldsymbol{\lambda}$, the variable part of the Lagrangian is

$$q_{\boldsymbol{\lambda}}(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^{\mathsf{T}}\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{a}(\boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{x}.$$

If $\boldsymbol{a}(\boldsymbol{\lambda}) \notin \operatorname{Range}(\boldsymbol{Q})$, symmetry gives $\operatorname{Range}(\boldsymbol{Q}) = \operatorname{Ker}(\boldsymbol{Q})^{\perp}$. Choose $\boldsymbol{z} \in \operatorname{Ker}(\boldsymbol{Q})$ with $\boldsymbol{a}(\boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{z} \neq 0$. Along $t\boldsymbol{z}$, the quadratic term vanishes and the linear term tends to $-\infty$ for one sign of t. Thus $\theta(\boldsymbol{\lambda}) = -\infty$.

Otherwise set $\boldsymbol{x}_0 = -\boldsymbol{Q}^{\dagger}\boldsymbol{a}(\boldsymbol{\lambda})$. The range condition gives $\boldsymbol{Q}\boldsymbol{x}_0 = -\boldsymbol{a}(\boldsymbol{\lambda})$. Expanding around $\boldsymbol{x}_0$ cancels the linear terms and yields

$$q_{\boldsymbol{\lambda}}(\boldsymbol{x}) = q_{\boldsymbol{\lambda}}(\boldsymbol{x}_0) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}_0)^{\mathsf{T}}\boldsymbol{Q}(\boldsymbol{x} - \boldsymbol{x}_0).$$

Positive semidefiniteness shows that the minimizer set is $\boldsymbol{x}_0 + \operatorname{Ker}(\boldsymbol{Q})$. Its minimum value is

$$-\frac{1}{2}\boldsymbol{a}(\boldsymbol{\lambda})^{\mathsf{T}}\boldsymbol{Q}^{\dagger}\boldsymbol{a}(\boldsymbol{\lambda}).$$

Adding the constant $-\boldsymbol{b}^{\mathsf{T}}\boldsymbol{\lambda}$ proves the formula. □

This calculation shows concretely why singularity need not mean unboundedness: a compatible linear term gives a finite minimum with a flat family of minimizers. An incompatible component along the nullspace gives an unbounded inner problem.

18.3.3 Supplement: constraint encoders and optimization order

To understand what changes when strong duality holds, we can express both optimal values using the same Lagrangian with a different order of operations. The starting point is the following scalar calculation.

Lemma 18.7 (Scalar constraint encoders)**.** *For a scalar t,*

$$\sup_{\lambda\in\mathbb{R}} \lambda t = \begin{cases} 0, & t = 0, \\ +\infty, & t \neq 0, \end{cases} \qquad \sup_{\nu\geq 0} \nu t = \begin{cases} 0, & t \leq 0, \\ +\infty, & t > 0. \end{cases} \tag{18.24}$$

Proof. If $t = 0$, every equality-multiplier term is zero. If $t > 0$, send $\lambda \to +\infty$; if $t < 0$, send $\lambda \to -\infty$. For the second formula, $t \leq 0$ makes every νt nonpositive, and $\nu = 0$ attains zero. If $t > 0$, sending $\nu \to +\infty$ makes the value arbitrarily large. □

Apply these formulas to each constraint at a fixed $\boldsymbol{x}$. If the point is feasible, then

$$\sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\, \boldsymbol{\nu}\geq \mathbf{0}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\boldsymbol{x}).$$

If any constraint is violated, the corresponding multiplier sends that supremum to $+\infty$, while the remaining multipliers can be set to zero. Therefore

$$p^* = \inf_{\boldsymbol{x}\in\mathbb{R}^n} \sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\, \boldsymbol{\nu}\geq \mathbf{0}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}). \tag{18.25}$$

The dual uses the opposite order:

$$d^* = \sup_{\boldsymbol{\lambda}\in\mathbb{R}^m,\, \boldsymbol{\nu}\geq \mathbf{0}} \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Thus strong duality is an equality between these two orders of optimization. It does not follow from a formal interchange of the symbols.

For completeness, take any real-valued function Φ on a product of two nonempty sets. For every fixed x, y,

$$\inf_u \Phi(u, y) \leq \Phi(x, y) \leq \sup_v \Phi(x, v).$$

The leftmost quantity is consequently bounded by the rightmost one for every x, y. Take a supremum over y, then an infimum over x:

$$\sup_y \inf_x \Phi(x, y) \leq \inf_x \sup_y \Phi(x, y). \tag{18.26}$$

This order inequality is another proof of weak duality when applied to L. Equality needs additional structure.

To see why smoothness and compactness alone are insufficient, take $\Phi(x, y) = (x - y)^2$ for $x, y \in [-1, 1]$. For fixed x, the largest squared distance to the interval is $(1+|x|)^2$, minimized at $x = 0$ with value 1. For fixed y, choosing $x = y$ makes the inner minimum zero. Hence

$$\inf_x \sup_y \Phi(x, y) = 1 > 0 = \sup_y \inf_x \Phi(x, y).$$

Chapter summary

- The dual function is the global infimum $\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$. It is concave even when the primal problem is nonconvex, but it can be nonsmooth. The dual maximizes it over unrestricted $\boldsymbol{\lambda}$ and nonnegative $\boldsymbol{\nu}$.
- A finite lower bound requires $(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{F}_D$. Primal unboundedness implies $\mathcal{F}_D =$

$\varnothing$ and $d^* = -\infty$; dual unboundedness above implies primal infeasibility. The converses need not hold.

- For a primal-feasible point and sign-admissible multipliers with finite θ, weak duality gives
$$\theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq d^* \leq p^* \leq f(\boldsymbol{x}).$$
Thus $G = f(\boldsymbol{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$ bounds the primal objective error, the dual objective error, and the optimal duality gap. Matching values $G = 0$ certify both global optimizers without convexity.
- Strong duality means $p^* = d^* \in \mathbb{R}$. Boundedness, attainment, and uniqueness must be checked separately for both the inner infimum and the outer supremum; zero gap alone does not supply an optimal pair.
- To recover a primal optimizer from a global Lagrangian minimizer, also check primal feasibility and complementary slackness. These turn its Lagrangian value into a matching primal objective; stationarity alone does not provide this certificate.
- At sign-admissible multipliers, an approximate inner value $L(\widetilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ is an upper bound on θ. If its global objective error is certified to be at most δ, subtracting δ gives a valid lower bound on p^*.

18.4 Exercises

Exercise 18.1 (Multiplier signs)**.** Rewrite $2 - x \geq 0$ in the course convention $h(x) \leq 0$, form its Lagrangian term, and explain the required multiplier sign.

Exercise 18.2 (A value certificate)**.** For $\min \frac{1}{2}(x_1^2 + x_2^2)$ subject to $x_1 + x_2 = 4$, evaluate the primal–dual gap at $\boldsymbol{x} = (1, 3)^\mathsf{T}$ and $\lambda = -3/2$. What objective-error bound follows? Does the Lagrangian minimizer at this multiplier satisfy the equality?

Exercise 18.3 (Inner attainment and effective domain)**.** Derive the dual of $\min e^x$ subject to $1 - e^x \leq 0$, distinguishing boundedness and attainment of every inner infimum. Then consider $\min -(x^2 - 1)^2$ subject to $x = 0$, and explain why the primal optimum exists although no finite dual value is available. Check that replacing this objective by $-x^4$ gives the same domain phenomenon, now with $p^* = 0$.

Exercise 18.4 (Primal recovery)**.** For $\min e^x$ subject to $1 - e^x \leq 0$, find the dual optimizer and all associated Lagrangian minimizers. Identify which are primal feasible and which satisfy complementary slackness. Explain why feasibility alone does not select the primal optimum.

Exercise 18.5 (A singular quadratic dual)**.** Let
$$\boldsymbol{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \boldsymbol{c} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \boldsymbol{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad b = 0.$$

Use Proposition 18.6 to determine the effective domain and dual function. Solve the primal and dual problems and compare their values.

Exercise 18.6 (A certified approximate inner value)**.** Suppose a fixed-multiplier Lagrangian is strongly convex with modulus $a > 0$. Show how an inner point with $\|\nabla_{\boldsymbol{x}} L\|_\infty \le \varepsilon_{\text{in}}^{(k)}$ produces a certified lower bound. Explain why the same argument cannot be used merely from a small gradient in a nonconvex inner problem.